Segmentation & Representation Benchmark Matrix

Consolidated validation across three corpora (MeerKAT, LibriSpeech, InfantMarmosetsVox), empirical

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Terminology — what the pipeline discovers in animal audio

Speech word-discovery methods (DUSTED, ES-KMeans, Bayesian word segmentation, SSL-unit clustering

speech (human, has GT)	animal (no GT meaning)
word	recurrent acoustic term / motif
word-like unit	motif-like acoustic term
phoneme / subword	sub-call acoustic unit
lexicon	motif inventory
sentence	bout / vocalisation sequence

All calltype/caller "enrichment" reported below is **acoustic/structural association, not semantics

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1. Event-detection F1 (which segmenter, which regime)

corpus (regime)	method	metric	F1	P	R	ref
MeerKAT (callsign noise, 8 kHz)	raw NBD	onset 1-to-1	0.069	—	—	massive over-segmentation
	sweep-best threshold	onset 1-to-1	0.165	—	—	thresholds insufficient
	v2 scalar GBT (19 feat)	onset 1-to-1	~0.26	—	—	tabular baseline (frozen)
	v3/v4 post-onset GBT	onset 1-to-1	~0.21	—	—	tabular **ceiling**
	PCEN+CRNN	onset 1-to-1	**0.703**	—	—	**8.9× raw NBD → gap closed**
LibriSpeech (scalar GBT speech, word-level)	scalar GBT speech, word-level	word 1-to-1 / combined	0.450 / 0.638	—	—	CPC-free
	CPC-context GBT (+dim)	word 1-to-1 / combined	0.458 / 0.794	—	—	CPC adds +0.156 combined
	NBD+CPC calibrated	**combined (§1 metric)**	**0.836**	0.77	0.92	**speech winner**
	PCEN+CRNN (retrained)	word 1-to-1 / combined	0.590 / 0.652	0.80	0.47	CRNN is the **speech bottleneck**
InfantMarmosetsVox (clean, adult)	**PCEN+CRNN (retrained)**	onset 1-to-1	**0.875**	**0.982**	0.789	**strongest result**

Router read: PCEN+CRNN wins for discrete calls in noise (0.70 meerkat → 0.88 marmoset); NBD+CPC

Tolerance sweep (apples-to-apples — `scripts/tolerance_sweep.py`)

The headline F1s used different tolerances (MeerKAT 100ms, marmoset 200ms). At **matched**

tolerance

onset F1	50ms	100ms	200ms
MeerKAT (500 clips)	0.662	0.707	0.724
InfantMarmosetsVox (30 test)	**0.864**	**0.865**	**0.867**

offset F1	50ms	100ms	200ms
MeerKAT	0.549	0.664	0.702
InfantMarmosetsVox	**0.785**	**0.860**	**0.866**

Marmoset onset F1 is tolerance-*insensitive* (0.864→0.867) because precision ≈ 0.98 (near-

zero false

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2. HECB internal-boundary mechanism (oracle-parent, no CRNN)

The internal mechanism decoupled from the detector — given valid parent events, does HECB

recover m

corpus	parent events	sub-event recovered	F1	P	R
LibriSpeech	oracle word spans	in-word phoneme start	**0.636**	0.781	0.537
LibriSpeech	CRNN word events	in-word phoneme start	0.409	0.516	0.339

Read: the mechanism works at 0.636 with good parents but degrades to 0.41 with poor

(CRNN) pa

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3. Whole / sub-call / pooled representation recovery

The core claim: sub-call features **augment** whole-event representations when pooled, but

are weake

Call-type recovery

corpus	split	whole V	pooled V	whole probe	pooled probe	whole macroF1	pooled macroF1
MeerKAT (end-to-end, 1500 clips)	—	0.246	**0.274**	0.611	**0.633**	—	—
InfantMarmosetsVox (end-to-end)	—	0.344	**0.379**	0.934	**0.940**	—	—
InfantMarmosetsVox (oracle)	random	0.431	0.416	0.935	0.932	0.487	**0.566**
InfantMarmosetsVox (oracle)	within-callid	0.477	0.475	0.944	**0.947**	0.557	**0.582**

*Sub-call alone is consistently weaker on V (loses whole-call context) but highest on probe

(sub-fragm

Caller-ID recovery (InfantMarmosetsVox oracle, random split, 10 callers) — **the novel finding**

representation	V	NMI	probe	macro-F1
whole call	0.178	0.070	0.459	0.364
whole + sub-call pool	**0.197**	**0.083**	**0.477**	**0.392**
sub-call	0.115	0.029	0.317	—

HECB internal boundaries recover caller identity — pooled wins on all four metrics.

Individual

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4. The pooled-adds-value thesis (cross-corpus summary)

corpus	label	pooled > whole?	on which metric
MeerKAT	call-type		V, NMI, probe (clean)
InfantMarmosetsVox (end-to-end)	call-type		V, NMI, probe
InfantMarmosetsVox (oracle)	call-type	mixed	macro-F1 (+0.079); whole edges V
InfantMarmosetsVox (oracle)	**caller ID**		**all 4 metrics**

Refined conclusion (post-controls): Sub-call structure augments whole-call

represent

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5. Segmentation router (canonical)

regime	method	HECB role	example
discrete calls in noise	PCEN+CRNN	internal annotator	meerkat (0.70), marmoset (0.88)
continuous speech-like	NBD+CPC calibrator	optional	LibriSpeech (0.836 combined)
exploratory / unknown	NBD + ABSTAIN	none	new species (conservative)

Updated core claim (project thesis): The pipeline uses a regime-specific segmentation

router: PC

Caveats

All F1s are one-to-one greedy matching (no duplicate-TP inflation). MeerKAT onset

toleranc

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6. Cross-species 4-way ablation (raw-NBD oracle, 2026-06-22)

`scripts/cross_species_ablation.py` — per-file oracle-parent HECB (raw NBD internal

boundarie

corpus (kHz)	label	whole V	pooled V	**sequence V**	whole probe	pooled probe	seq probe
**adult marmoset (vocalization, 96)	call-type (6)	0.458	0.434	**0.468**	0.824	0.832	**0.837**
**egyptian fruit bat (250)	Context (2)	0.152	0.152	0.153	0.547	0.556	0.543
egyptian fruit bat (250)	Emitter (ID)	0.291	0.291	0.291	0.211	0.191	0.195

Honest read — this REFINES the pooled-adds-value thesis:

**The re

Revised cross-species claim: sub-call structure adds value where (a) the internal

boundarie

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7. Typed HECB vs raw NBD on adult marmoset (boundary-quality test, 2026-06-22)

`scripts/typed_hecb_vs_raw.py` — same eval files (2491 calls, 6 types), two internal-

boundary
lacked.

method	whole V	pooled V	pooled−whole	pooled probe	sub-call V	sub-call probe
raw NBD	0.450	0.418	−0.032	0.800	0.173	0.487
typed HECB	0.450	**0.431**	**−0.019**	**0.814**	**0.255**	**0.527**

Honest result: typed HECB improves sub-call fragment quality (V 0.255 vs 0.173, probe

0.527 vs (

Cross-experiment pattern (now consistent): the sub-call advantage appears on

**supervis

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8. Caller-ID confound controls (InfantMarmosetsVox, 2026-06-22)

`scripts/caller_id_controls.py` — validates the novel caller-ID finding against confounds.

2828 vocs

Baseline + within-call-type (V / NMI / probe)

condition	whole V	pooled V	pooled−whole V	whole probe	pooled probe
all call types	0.185	**0.203**	+0.018	0.452	**0.479**
within Phee (10 callers)	0.223	**0.234**	+0.011	**0.529**	0.465

within Twitter (10 callers)	0.251	**0.261**	+0.010	0.594	**0.601**

Trivial baselines (probe accuracy)

baseline	probe acc	macro-F1	interpretation
duration-only	0.165	0.070	near-chance (0.10) — **NOT duration**
calltype-onehot	0.138	0.059	near-chance — **NOT call-type confound**
whole-fp (dur-regressed)	0.435	0.393	survives duration removal

Verdict: the caller-ID finding is **genuine** — it survives all confound controls.

The poole

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9. Track C bat context ablation (acoustic + social + behavioral, 2026-06-22)

`scripts/bat_trackc_ablation.py` — the scientific gap: does combining modalities recover

context b

Context recovery — the A–G pairwise ablation matrix (the key target)

model	features	probe acc	macro-F1	lift
A	acoustic-only (fingerprint)	0.249	0.218	2.7×
B	social-only (emitter+addressee)	0.267	0.223	2.9×
C	behavior-only (pre/post actions)	0.087	0.162	1.0×
D	acoustic + social	**0.393**	**0.341**	**4.3×**
E	acoustic + behavior	0.298	0.307	3.3×
F	social + behavior	0.369	0.327	4.1×
G	acoustic + social + behavior	**0.424**	**0.355**	**4.7×**

The pairwise decomposition reveals where the synergy lives:

**D (aco

Leakage / generalization controls (reviewer-proof)

split	A acoustic	B social	D acous+social	G full
random	2.7×	2.9×	**4.3×**	**4.7×**
social-shuffle (emit/addr permuted)	2.7×	**1.0×**	**3.2×**	4.0×

emitter-held-out (GroupShuffleSplit)	2.6×	2.8×	**4.1×**	**4.8×**

Social-shuffle collapses B to chance (1.0×) and drops D by 1.1× (4.3→3.2) → ~69% of D's lift o

Manuscript-ready Track C conclusion

Acoustics encode call form and caller-associated structure, but ethological context requires *

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10. Sub-call motif structure report (InfantMarmosetsVox, 2026-06-22)

`scripts/subcall_motifs.py` + `scripts/motif_controls.py` — 2423 vocalisations, k=12 fingerprin

Position-specialized clusters (onset/body/offset roles)

cluster	first%	middle%	last%	**role**	calltype V	caller V
7	**84.0**	15.7	0.2	**onset**	—	—
1	0.6	4.2	**92.4**	**offset**	—	—
0	2.3	**97.6**	0.1	**body**	—	—
10	0.0	**100.0**	0.0	**body**	—	—
11	0.2	**99.8**	0.0	**body**	—	—
3	29.7	**68.3**	2.0	body (onset-leaning)	—	—

Calltype-enriched motifs (top by V)

level	motif	count	calltype V	p
1-gram	cluster 0	1408	0.653	<0.0001
1-gram	cluster 7	1092	0.650	<0.0001
2-gram	(0, 0)	462	0.398	<0.0001
2-gram	(0, 7)	307	0.372	<0.0001
3-gram	(11, 11, 11)	175	0.242	<0.0001
3-gram	(0, 7, 0)	104	0.227	<0.0001

Control summary

control	result
bootstrap stability (bigram Jaccard)	**1.000** (STABLE)
bootstrap stability (trigram Jaccard)	**0.773** (STABLE)
position enrichment	onset/body/offset roles confirmed (84%/9
within-Phee caller motifs	(10,0) V=0.186 p<0.0001 — caller signal
bigram-preserving trigram null	374 vs 381±5 (5th pct) — trigrams ≈ bigr
token-shuffle bigram richness	94 vs 104±1 — **non-random** (constraine

Manuscript-ready claim

HECB-derived sub-call units are stable, non-random, and positionally specialized.

Cluster 7

11. DUSTED-style recurrent-subsequence motif discovery (3-way method comparison, 2026-06-22)

Adapts spoken-term-discovery (DUSTED) to **recurrent acoustic terms** in animal audio

(`diagnos

exact n-grams — windows match iff cluster-ID tuples are identical (fragments every

substitut

Run: `python scripts/hecb_dusted_marmoset.py --n-eval 1500` (`scripts/hecb_dusted_marmoset.py`).

Method comparison (the decisive question — variant tier, DTW thr=0.35)

method	#motifs	total support	coverage	mean calltype V	mean caller V
exact n-grams	376	6236	0.155	0.109	0.102
fixed-length cosine	18	5988	0.509	0.289	0.125
DTW with indels	**2**	**6970**	**0.516**	**0.508**	**0.170**

DTW merges variants into the fewest, richest families: highest support/coverage and

**calltype

Honest caveats (do not overclaim)

1. **Degenerate inventory.** InfantMarmosetsVox sub-call structure is dominated by ONE

recurren

2. **Order-shuffle null is conservative for repeats.** The within-call order shuffle

preserv

3. **Bootstrap Jaccard low (0.16–0.18).** With a degenerate inventory (1 dominant + sparse minors)

Manuscript-ready claim

DTW-with-indels is the right subsequence-matching primitive for recurrent-acoustic-term discovery

12. Bengalese-finch song — the rich-repertory positive control (GT-validated, 2026-06-22)

The marmoset caveat (§11) predicted a richer inventory from a richer corpus. Bengalese finch song

3-way method comparison (mean across Bird0/1/2)

method	#motifs	coverage	label purity (mean / wmean)
exact n-grams	106	0.84	1.00 / 1.00
fixed-length cosine	104	0.90	0.97 / **0.895**
DTW with indels	81	0.90	0.83 / 0.59

The expected, clean method trend — and crucially, **fixed-length cosine recovers the GT syllable m

DTW recovers real syllable motifs blind from acoustics (selected, per bird)

bird	motif (GT labels)	support	purity	note
Bird1	`[1,0]`	304	**0.90**	clean 2-syllable motif across 141 cal
Bird2	`[4,5]`	158	**1.00**	perfectly recovered bigram, 118 calls
Bird2	`[5,6]`	148	0.78	—
Bird2	`[10,11,12,13]`	209	0.51	4-syllable motif
Bird0	`[2,3,4]` / `[6,2,3]` / `[5,6,15]`	6–10	0.51–0.56	trigrams

Nulls + stability (per bird)

Observed DTW support is **2.0–2.3× the order-shuffle null** on every bird (p=0.111 is the

n_perm=8

Manuscript-ready claim

On Bengalese-finch song — the right dataset class for diverse recurrent acoustic terms —

DUSTED c

13. Egyptian fruit bat — emitter/social context; graded, not discrete (2026-06-22)

The stress test the Bengalese result (§12) set up: does a *socially annotated* corpus

form recu

The decisive finding: Emitter V ≫ Context V (every method)

method	#fams	coverage	H(norm)	top-frac	**Emitter V**	**Context V**
exact n-grams	582	0.29	0.91	0.04	0.463	0.164
fixed-length cosine	155	0.78	0.60	0.43	0.454	0.167
DTW with indels	105	0.82	0.32	0.72	0.467	0.163

**Emitter V ≈ 0.45–0.47 vs Context V ≈ 0.16–0.17 (~2.8×) — rock-solid across all three

methods.*

Bat is graded/intermediate, not discrete-syllable like Bengalese

The diverse-vs-degenerate diagnostic (the user's `family_entropy` / `top_family_support_fraction`):

bat sits **

Honest caveats

Null modest (obs/null ≈ 1.23×, p=0.2 at n_perm=4): the dominant `[12,12]` repeat is

composi

Manuscript-ready claim

On Egyptian fruit bat, recurrent acoustic-term families are tied to **emitter/caller

(0.45–0.4

14. Unsupervised word-structure benchmark — HECB vs SSL discrete units (LibriSpeech, 2026-06-22)

The decisive external comparison: is HECB's boundary-aware internal sub-unit sequence

competiti

Headline table

unit source	within	across	a_dur	a_phn	ratio	NN acc	NN_xspk	motif→word V
HECB	0.908	0.930	0.939	0.929	0.976	0.089	0.085	0.326
HuBERT-km(50)	0.928	0.920	0.924	0.923	**1.008**	0.052	0.052	0.302
CPC-km(50)	0.900	0.943	0.944	0.943	**0.954**	**0.191**	0.178	0.322
fingerprint(328)	0.897	0.931	0.934	0.933	0.964	0.147	0.142	**0.347**

Controls (all clean)

`a_dur` / `a_phn` (across-word baseline matched on duration / phoneme-count) ≈ `across`

for every

Findings

1. **CPC-km wins** raw same-word consistency (ratio 0.954, NN 0.191). The pipeline's own

SSL mo

Honest caveats (do not overclaim)

HuBERT's failure is metric-specific, not a defeat of HuBERT as a representation.

The zero

Manuscript-ready claim

On a controlled LibriSpeech same-word-consistency benchmark, HECB's boundary-aware

sub-unit s

> **Fingerprint note (all experiments):** the `fingerprint` source is `afp.extract(...).combined`

> = the
species-n

15. Stage C + D — DUSTED term discovery & ES-KMeans clustering (HuBERT under DTW; LibriSpeech, 2026-06-23)

Two follow-on stages closing the §14 "HuBERT fails" loop (was it the metric, not the unit?).

Both reus

Stage C — DUSTED word-term discovery (DTW, HuBERT's native metric; `dusted_librispeech_words.py`)

Each word instance = one CallUnits (discrete-unit sequence, `calltype=word`). DUSTED module run

per source

unit source	exact word_V	DTW word_V	DTW fams	DTW cov	top-frac

HECB(k32)	0.409	0.425	254	0.048	0.02
HuBERT-km(50)	0.415	0.418	501	0.122	0.02
CPC-km(50)	0.418	0.431	651	0.075	0.01
fingerprint(32)	**0.448**	**0.458**	259	0.078	0.03

HuBERT recovers under DTW: word_V 0.418 (vs §14's Levenshtein *failure* ratio 1.008). So

§14's "Hu

Stage D — whole-word embedding clustering (ES-KMeans lexicon core; `eskmeans_librispeech.py`)

Oracle GT word spans → whole-word acoustic embedding → k-means (k=60) → evaluate vs GT word

labels. Co

embedding	V-measure	ARI	purity	token-F1
fingerprint (44-D)	**0.481**	0.061	**0.276**	**0.276**
CPC-mean (256-D)	0.436	0.036	0.242	0.242
HECB-pooled (44-D)	0.392	0.012	0.198	0.198
HuBERT-mean (768-D)	0.338	−0.003	0.172	0.172

Fingerprint clusters words best (V=0.481, token-F1 0.276 = 1.6× chance over 88 types);

HuBERT-m

Manuscript-ready claim

Across four complementary LibriSpeech word-structure tests (§14 Levenshtein, Stage C DTW term

discovery

16. HuBERT sanity ablation — the conclusion is locked (2026-06-23)

The reviewer-safe sweep: does ANY HuBERT configuration (layer × k × collapse × pooling)

beat the f

Track 1 — sequence consistency (18 configs: layer ∈ {6,9,12} × k ∈ {50,100,200} × {col,raw})

Every single HuBERT config has ratio **1.001–1.018** (within ≥ across: *no* same-word structure)

Track 2 — whole-word clustering (6 configs: layer ∈ {6,9,12} × pool ∈ {mean,stat})

Best HuBERT: **L6 stat (V=0.331, tokF1=0.207, ARI=−0.006)**. Fingerprint REF: V=**0.517**, tokF1=**(

Locked conclusion

Across **24 HuBERT configurations** spanning the standard zero-resource design space (layers 6/9/12, k

17. Bat held-out emitter validation — the claim generalizes (2026-06-23)

The §13 claim (bat recurrent acoustic-term families track emitter identity ~2.8× more than context) v

metric	Train (70%)	Held-out (30%)
Emitter V	0.221	**0.375**
Context V	0.085	0.149
ratio	2.60×	**2.52×**

85.5% of held-out windows assigned; 98.5% bout coverage. Held-out assignment is **diverse** (entropy

18. RookID individual-ID — whole-call fingerprint wins, extends bat emitter finding (2026-06-23)

The headline RookID experiment: does hierarchical sub-call structure improve individual identificat

representation	Source V	NMI	macro-F1	#train	#test	#src
whole-call fp (44-D)	**0.384**	**0.320**	**0.287**	2055	944	14
HECB-pooled fp (44-D)	0.321	0.242	0.263	2055	944	14

Findings:

1. Whole-

Dataset summary: 15,008 calls (excl Inc/Pls), 15 named rooks, 514 events (many "sing"

variants –

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19. Clusterer robustness of the pooled representation — k-means is NOT the confound (2026-06-23)

`scripts/marmoset_clusterfn_compare.py` — answers a question raised in §6/§7: is the weak

pooled-vs

clusterer	pooled V	pooled NMI	whole V	sub V	probe (whole)
k-means (spherical)	**0.440**	**0.244**	0.409	0.177	0.820
cosine-HAC	0.277	0.113	0.275	0.249	0.820
GMM (diag)	0.335	0.146	0.394	0.181	0.820

The architectural hypothesis is FALSIFIED. k-means gives the *highest* pooled V/NMI of

the three,

1. **The pooled augmentation is real and clusterer-robust** for the leading clusterer

(k-mean

This refines the §7 cross-experiment pattern: the sub-call advantage shows on supervised metrics (p

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20. Track C CCA — acoustic ↔ social share a genuine common subspace (2026-06-23)

`scripts/bat_trackc_cca.py` — unsupervised dual-validation of the §9 supervised 4.7× context-re

canonical pair	ρ	ρ²
ρ_1	**0.771**	0.595
ρ_2	0.699	0.488
ρ_3	0.623	0.388
ρ_4	0.599	0.359
ρ_5	0.541	0.293

total R^2 (first 5 comps) = 2.123

permuta

The Track C multimodal synergy is intrinsic, not a classifier artifact. A leading canonical

Honest caveat. The permutation p-value on ρ_1 is marginal (0.045), and the social space is high-dir

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21. LibriSpeech UNIFIED rerun — same words for §14 + Stage C + Stage D (2026-06-23)

`scripts/ls_unified_rerun.py` — fixes the apples-to-apples gap: the three LS word-structure scripts (§1

§14 — within/across edit-dist + NN + motif→word (n=600)

source	within	across	ratio	NN	NN_xsp	motifV
HECB(k32)	0.917	0.935	0.981	0.098	0.090	0.429
HuBERT-km(k50)	0.939	0.928	**1.012**	0.047	0.047	0.424
fingerprint(k32)	0.907	0.942	0.964	0.118	0.114	**0.481**
CPC-km(k50)	0.900	0.940	0.958	**0.175**	**0.171**	0.458

Stage C — DUSTED recurrent-term discovery (DTW row, same n=600)

source	#fams	cov	word_V	H	top-frac	null-ratio	bootstrap
HECB(k32)	260	0.038	0.419	0.98	0.02	1.15×	0.181
HuBERT-km(k50)	617	0.080	0.416	0.96	0.01	1.35×	**0.428**
fingerprint(k32)	269	0.068	**0.453**	0.96	0.03	1.26×	0.298
CPC-km(k50)	738	0.052	0.427	0.97	0.01	1.21×	0.310

Stage D — whole-word embedding clustering (same n=600, k=60)

source	V	ARI	purity	tokenF1
fingerprint	**0.517**	0.054	0.282	0.282
HECB-pooled	0.451	0.013	0.223	0.223
HuBERT-mean	0.386	−0.003	0.193	0.193
CPC-mean	0.463	0.033	0.248	0.248

Unified verdict (the point of putting all three on the same words)

The fingerprint leads on every aggregate metric: §14 motifV 0.481, Stage-C DTW word_V

0.453, S

Run: `python scripts/ls_unified_rerun.py --n-files 200 --max-words 600`; re-score:

`python s

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22. Cross-corpus DUSTED motif table — uniform threshold, HAC-corrected (2026-06-23)

The three animal DUSTED scripts (`hecb_dusted_marmoset`, `dusted_bat`, `dusted_bengalese`) now

emit iden
tier)**

corpus (n)	#fams	cov	purity(w)	H(norm)	top-frac	null-ratio	bootstrap
Bengalese finch (3 birds, 1	457	0.480	**0.788**	0.85	0.12	**2.67×**	**0.509**
Egyptian fruit bat (300 bouts)	306	0.153	0.39 †	0.87	0.06	1.36×	0.394
Infant marmoset (1500 calls)	139	0.028	0.49 †	0.89	0.06	1.55×	0.229

† purity label is corpus-specific (bengalese = GT syllable sequence → genuine blind recovery;

bat = Cor

The cross-corpus claim holds on honest (HAC-corrected) numbers. The relative ordering —

Bengalese

§11 marmoset: "DTW ct_V 0.508, 4.7× exact" was a CC-chaining artifact — HAC-corrected

marmo

call-type

Manuscript framing: DUSTED lands exactly where the repertoire predicts — diverse discrete

syllables

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23. RookID Event/Comment enrichment — whole-call fp vs DUSTED motifs × 6 targets (2026-06-23)

`scripts/rookid_event_comment.py` — keeps the two columns the prior RookID scripts dropped:

**`Event`

target	n	#cls	fp-V	fp-NMI	fp-probe	motif-V	motif-NMI
Source (individual)	2798	14	0.308	0.175	0.325	0.347	0.156
activity (singing/call)	2730	2	0.673	0.159	**0.846**	0.657	0.092
sing/non-sing	2730	2	0.673	0.159	**0.846**	0.657	0.092
song_variant (song-prefix)	1938	14	0.242	0.087	0.118	0.327	0.129
social_other (partner rook)	2809	2	**0.041**	0.001	n/a	0.176	0.001

date (recording)	2809	4	0.249	0.055	n/a	0.315	0.057

Findings (a clean individual-vs-social separation):

1. **Indivi

Manuscript framing: across RookID targets the pattern is *what* (activity/sing) and *who*

(Source) a

---

24. Soundwel Track C — does acoustic structure predict emotional valence beyond call type? (2026-06-23)

`scripts/soundwel_trackc.py` — Soundwel = 6887 labelled pig vocalizations (44.1 kHz,

~0.4 s) w

6-feature × 3-target probe (Team-held-out macro-F1)

target	whole fp	HECB	seq	motif	ctype	whole+ctype
vocalisation_type (chance 0.17, k=6)	**0.417**	0.275	0.382	0.288	—	—
production_context (chance 0.50, k=2)	0.662	—	—	—	**0.790**	0.626
emotional_valence (chance 0.50, k=2)	0.478	—	—	—	0.296	0.466

Decisive valence block (the "beyond call type" test; chance 0.50)

test	macro-F1
call-type → valence (baseline)	0.508
fp → valence	0.478
fp + call-type → valence	**0.466** (adding fp *hurts*)
within-call-type fp → valence	**0.350** (well below chance; 6 calltype
fp → valence (male / female)	0.339 / 0.352 (both below chance)

Verdict — the bioacoustic semantic ceiling, proven

**Acoustic structure does NOT predict emotional valence — it is at/below chance, including

the clean

Manuscript framing: emotional valence is locked inside the call-type template; it is not linearly se

---

25. LibriSpeech speaker-invariant / universal acoustic-structure test (2026-06-23)

`scripts/ls_speaker_invariant.py` — does the ordered acoustic sub-unit representation preserve

4-way distances (ratio = d(diff-word,same-spk) ÷ d(same-word,diff-spk); >1 ⇒ word > speaker)

representation	sw/ds	dw/ss	ratio
fp_whole (cosine)	0.160	0.137	**0.86**
fingerprint-km	0.904	0.886	0.98
HECB	0.904	0.901	1.00
CPC-km	0.900	0.875	0.97
HuBERT-km	0.929	0.908	0.98
HuBERT-DTW (raw-frame, n=300)	0.864	0.826	0.96

Cross-speaker same-word retrieval (eval=400, gallery=1895; gallery excludes query speaker; chance≈0.015)

representation	top-1	top-5	MRR
CPC-km	**0.105**	**0.315**	**0.212**
fingerprint-km	0.087	0.233	0.163
fp_whole (cosine)	0.083	0.230	0.162
HECB	0.068	0.185	0.135
HuBERT-DTW (n=300)	0.042	0.083	0.074
HuBERT-km (Levenshtein)	0.018	0.098	0.062

Word-vs-speaker disentanglement + centering (fp_whole)

word probe (speaker-held-out macro-F1) = **0.110**; speaker probe = 0.094 (chance 0.006).

word pro

Honest verdict — a *qualified* universal representation

1. **Cross-speaker same-word retrieval is well above chance** (CPC 7×, fp/fingerprint ~5–6×):

word-as

Manuscript-safe claim: *cross-speaker same-word retrieval from ordered acoustic sub-unit

represent

---

26. RookID cross-Caller same-call retrieval — the §25 animal analogue (2026-06-23)

`scripts/rookid_cross_caller.py` — the cross-species mirror of §25: does CALL-TYPE structure

survive ac

target (call-type	n label)	#labels	chance	fp top-1	fp top-5	fp MRR	ratio
activity (sing/ca	2809	2	0.572	0.665	0.910	0.771	1.00
sing/non-sing	2809	2	0.572	0.598	0.882	0.719	0.97
song_variant (sing prefix)	2066	7	0.330	**0.403**	0.677	0.514	0.99
Event (fine code)	1677	30	0.240	**0.160**	0.393	0.273	0.98

(sub-call-seq Levenshtein is within ±0.02 of fp on every cell — both reps agree.)

Cross-species verdict — animal calls are MORE caller-bound than human words are speaker-bound

1. **Coarse category barely generalizes across rooks**: activity/sing cross-caller top-1

(0.60–0.

The decisive cross-species contrast (§25 ↔ §26): on LibriSpeech, cross-**speaker**

same-**w

Manuscript framing: *cross-speaker word retrieval is well above chance, but cross-caller

call-type

---

27. DTW-barycenter consensus word template — the §25 follow-on (negative) (2026-06-23)

`scripts/ls_consensus_template.py` — §25 found linear speaker-centering **fails** (Δ −0.003),

so the pri

method	top-1	chance
raw cross-speaker NN (DTW on fp-traject	**0.212**	0.040
DBA consensus template	**0.172**	0.040
Δ(template − NN)	**−0.040**	

Verdict — no simple consensus "universal word" (honest negative)

The DBA consensus template does **not** beat raw cross-speaker NN — it is slightly *worse*

(0.172 <

Manuscript framing: *there is no simple consensus "universal word" template derivable

from the

CPC-space replication — the representation matters (the sign flips)

Re-running with `--rep cpc` (continuous 256-D CPC frame trajectories, §25's most

speaker-i

representation	raw cross-speaker NN	DBA consensus template	Δ (template − NN)
fingerprint (43-D windows, phy	0.212	0.172	**−0.040**
HuBERT (768-D frames, MLM)	0.092	0.044 (≈chance)	**−0.048**
AVES (768-D frames, multi-tax	0.096	0.028 (below chance)	**−0.068**
**CPC (256-D frames, contrast	**0.428**	**0.440**	**+0.012**

The sign **flips** across representations: in the physics-bound fingerprint, DBA averaging

**smears

Honest magnitude. The effect is real but **modest** (+0.012, paired over the same 250

queries):

Firewall / animal translation. CPC is human-trained and firewall-blocked on animals, so

this speak

---

28. RookID sing/song-bout DUSTED — within-song motif inventory (2026-06-23)

DUSTED DTW discovery on the **sing subset** (1938 calls, 6 sources, 14 song variants)

of the cac

stat	value

#families	**1398**
H(norm)	**0.93** (diverse, non-degenerate — unli
top-family-frac	**0.01** (no dominant)
src_V (Source)	0.042
var_V (song_variant)	0.084
null ratio	1.06× (p=0.111 — composition-driven)
bootstrap Jaccard	0.341 (moderate)

representation	Source V	Source F1	song_variant V	Event V
whole fp (44-D)	**0.294**	**0.826**	**0.230**	**0.243**
HECB mean (44-D)	0.302	0.822	0.206	0.215
motif hist (50-D)	0.110	0.446	0.108	0.136

Findings: (1) Rook song HAS diverse recurrent acoustic structure (1398 families, H=0.93,

top=0.01

---

29. Cross-analysis: why recurrent acoustic terms are not decoded words (2026-06-23)

Five summary cross-cuts across the validated results (§1–§28), assembled into the

reviewer-

Cross-analysis 1 — cross-speaker word invariance vs cross-caller call-type invariance

Corpus	Unit tested	Identity held out	Retrieval target	Result	Interpretation
LibriSpeech	word	speaker	same word	**5–7× chance** (CPC 0.105 fa 0.03	words generalize across speaker
RookID	Event (fine code)	caller (Source)	same Event	**below chance** (0.160 vs 0.240)	fine calls are caller-bound
RookID	song_variant	caller	same variant	**moderate** (0.403 vs 0.330)	song structure partly shar
Egyptian bat	context	emitter	same context	expected weak (§13 V=0.16)	context not acoustic-invariant
InfantMarmosetsVox	call-type	caller/twin	same call-type	**strong** (V=0.606, §18)	call type is acoustic (form), not s

The decisive contrast: human **words** generalize across speakers (5–7× chance, §25);

animal **

Cross-analysis 2 — identity / call-form / context disentanglement across corpora

Dataset	Identity signal	Call/form signal	Context/social signal	Best representation
Egyptian bat	**high** (Emitter V≈0.45)	moderate (motifs graded)	**weak** (Context V≈0.16)	fingerprint; social needed for context
RookID	**high** (Source V=0.31)	ing/call strong (V=0.67)	**weak** (social_other V=0.04)	whole-call fingerprint
InfantMarmosetsVox	moderate (caller V present)	**strong** (call-type V=0.606)	receiver-side context	fingerprint + HECB roles
Bengalese finch	bird-constant (per-bird)	**strong** (GT syllable purity 0.78)	motif/acoustic context	exact/fixed-cosine motif
Soundwel pig	— (no individual ID)	call-type acoustic (0.41)	**valence not recoverable** (0.35 < chance)	fingerprint/CPC
LibriSpeech	speaker entangled (§25 ratio=1)	**strong structure** (motif V=0.48, §21)	word GT (controlled)	fingerprint/CPC

The pattern: animal acoustic structure carries *who called* and *what call form* more

reliably th

Cross-analysis 3 — word-likeness / lexicality stress test

Criterion	LibriSpeech	Bengalese	RookID	Bat	Marmoset
Cross-individual invariance	**high** (§25 5–7×)	per-bird (n/a)	**low** (§26 ≈chance)	low (§13 emitter-bound)	moderate (V=0.60 but within-ses
Motif inventory diversity	high	**high** (§12 H=0.85)	**high** (§28 H=0.93)	moderate (§13 graded)	**sparse** (§11 degenerate)
Label purity / enrichment	high (§21 motif V 0.48)	**high** (§12 0.79)	**low** (§28 V<0.1)	low (§13 V<0.17)	moderate (§18 V=0.61)
Context-independence	**high** (word GT)	n/a	**low** (§23 social≈0)	**low** (§9 social needed)	n/a
Overall word-likeness	**high**	moderate (acoustic, not semantic)	**low–moderate**	**low**	**low**

Reviewer-safe sentence: *The pipeline recovers structured recurrent acoustic terms, but

animal co

Cross-analysis 4 — evidence-tier status (H0-CMC ladder)

Claim	Tier	Evidence	Status
Recurrent acoustic terms exist	**T0**	DUSTED discovery (§11–§13, §22, §28)	validated
Individual identity recoverable	**T1**	bat Emitter V, marmoset V=0.61, RookID Source V	validated
Call-form recoverable from acoustics	**T1**	marmoset Soundwel call-type 0.41	validated
Context is socially grounded (not T1 output)	**T2+**	bat A–G D≈G=4.3×, survives shuffle-null	validated
Valence/emotion not acoustic	**T1**	Soundwel within-call-type 0.35 < caller (§	validated (ceiling)
Receiver detects motif structure	**T2**	—	**needs playback**
Multimodal receiver convergence	**T3**	—	**needs playback**
Causally supported functional signal	**T4**	—	**needs counterfactual playback**
Ecologically validated function	**T5**	—	**needs field replication**

The pipeline has exhausted T0–T1 (acoustic association + contextual prediction). T2–T4

(receiver

Cross-analysis 5 — playback-candidate prioritization

Dataset	Best playback candidate	Why	H0-CMC prediction
RookID	Source familiarity (known vs novel cal	clean individual labels; tests caller-re	response follows caller identity, not mo
InfantMarmosetsVox	onset/body/offset role-templated calls	mechanistic HECB role test (T2)	response disrupted by role-swap → roles
Egyptian bat	emitter-associated motifs vs context-soc	tests social grounding (T2); D=4.3× pro	response follows emitter, not context
Bengalese finch	GT motif order-shuffled vs natural	positive control (T2); GT syllable sport	response disrupted by order-shuffle → mo

Counterfactual controls (H0-CMC §4): natural / motif-preserved / order-shuffled /

caller-swa
amplitude

30. Boundary-quality sensitivity sweep — at what parent quality does hierarchy help? (2026-06-23)

Question (Architecture v1.0 invariant #2/#3). HECB annotates typed internal sub-call

boundarie

Setup. `hecb_decode` accepts any parent events, so parent quality is a clean independent variable;

MeerKAT (the HECB corpus): brief calls in noise + trained CRNN ⇒ parent quality varies via REAL de

MeerKAT — real CRNN detector operating points

method	parent F1	whole V	pooled V	gap	whole probe
oracle_typed	1.000	0.364	0.386	**+0.021**	0.753
oracle_raw	1.000	0.364	0.385	+0.021	0.753
crnn0.9_typed	0.940	0.299	0.351	+0.052	0.709
crnn0.7_typed	0.934	0.320	0.341	+0.022	0.645
crnn0.7_raw	0.934	0.320	0.332	+0.012	0.645
crnn0.4_typed	0.894	0.278	0.367	+0.089	0.640
raw_nbd	0.101	0.286	0.286	**0.000 ← abstain**	0.583

Marmoset — oracle/jittered parent (rich internal structure)

method	parent MAE (ms)	whole V	pooled V	gap	whole probe
oracle_raw	0.0	0.524	0.492	**−0.032**	0.810
jitter25_raw	12.9	0.522	0.486	−0.035	0.808
jitter75_raw	38.8	0.537	0.483	−0.054	0.804
jitter150_raw	77.6	0.524	0.487	−0.037	0.792
oracle_none (no internal)	0.0	0.524	0.524	0.000	0.810

Verdict — whole-call primary (invariant #3) confirmed on both corpora

Sub-call internal structure is NOT a call-type-recovery lever. At the best parent quality the hierarchy (pooled is

Consequence for the HECB gate: there is **no meaningful abstain threshold for hierarchy benefit**, because t parent).

Honesty caveat. The ablation V uses k=12 for sweep-internal consistency, so absolute whole V

(0.364 m
(meerkat

31. Cross-species repertoire taxonomy — WHY each corpus shows the structure it does (2026-06-23)

Question (Architecture v1.0 Step 3). The boundary sweep (§30) showed whole-call primary holds

regardless

corpus	repertoire type	secondary flags	null ratio	bootstrap	coverage	emitter V	context V	social lift
Bengalese finch	`discrete_syllable`	—	2.67×	0.509	0.480	n/a	n/a	n/a
Egyptian fruit bat	`graded`	identity_dominant	1.36×	socially_grounded	0.394	0.153	0.446	0.167	1.74
Infant marmoset	`sparse_repetitive`	—	1.55×	0.229	0.028	n/a	0.599	n/a
Rook	`individually_identity_dominant`	—	n/a	n/a	n/a	0.31	0.04	n/a
LibriSpeech (human)	`human_lexical`	—	—	—	—	—	—	—

Numbers curated from validated sections (§22 cross-corpus DUSTED; §13/§18/§23 binding V; §9 social-lift),

each cite
re-derive

Classifier (calibrated so each reference corpus lands on its validated type):

1. human

Two honest design notes. (1) DUSTED family-entropy is near-max for ALL corpora (0.85–0.89) — a

weak
emitter-vs

32. Identity–form–context disentanglement — what acoustics encode / don't / need-social (2026-06-23)

Question (Architecture v1.0 Step 4). The backbone biological table: for each corpus, what do the

acoustics
partner,

corpus	individual_id	call_type	context	social_partner
Egyptian fruit bat	**strong** (V 0.446)	weak (0.167)	**needs_social** (§9 D=4.3×)	n/a
Infant marmoset	n/a (no ID labels)	**strong** (V 0.599, probe 0.829)	n/a	n/a
MeerKAT	n/a	**strong** (V 0.521, probe 0.753)	n/a	n/a
Rook	partial (Source V 0.31)	**strong** (sing/call 0.67)	weak (social_other 0.04 = ≈ chance §26)	n/a (encode)
LibriSpeech (human)	**strong (speaker probe 0.56)**	partial (token↔word 0.29)	n/a (read speech)	n/a

The three-bucket thesis, operationalised:

**ENCC
meerkat,

Human anchor (the form/meaning split). Speaker identity IS strongly encoded (CPC probe 0.56 — the

latent is s
tokens

Steps 1–4 (the paper-ready core) are complete.

33. Structural alternatives ablation — DUSTED vs RQA / Matrix Profile / PST / Transition Network (2026-06-23)

Question. Four structure-discovery techniques that do NOT rely on DUSTED's token-sequence

alignment
Modules i

dataset (regime)	alternative	result	verdict vs DUSTED §22
Bengalese (discrete, diverse)	PST + Transition Network	PST **247** predictive contexts, depth 8; **639** nodes, 1205 edges, **19** motif bouts	**CORROBORATES** (purity 0.788); DTW's
Egyptian fruit bat (graded, n=26)	RQA + Matrix Profile	RQA **DET 0.229 / LAM 0.169**; MP **recurrence_fraction 0.788**, 1 motif	**COMPLEMENTS** — quantifies the grade
Infant marmoset (sparse)	Matrix Profile (+ RQA)	MP **recurrence_fraction 0.788**, 1 motif	**FAST PRE-FILTER** — token-free O(n log
LibriSpeech (diverse syntactic)	Transition Network (word transitions)	**639** nodes, 1205 edges, **19** motif bouts	**UPPER BOUND** — scales to a far larger

The takeaway — no alternative replaces DUSTED; each finds its regime niche. PST/Transition-Network

*corrobor
*complem

Honest caveats. (1) The wav2vec2 aligner emits word IDENTITIES + phoneme TIMES (not phoneme

identities)
the user

34. Three targeted structural reanalyses — closing DUSTED

limitations (2026-06-23)

The §33 alternatives applied surgically to three known DUSTED limitations — NOT a broad reanalysis.

Scripts: `scripts/{ `.

#1 Bat — RQA laminarity is context-dependent (the dynamical lens DUSTED lacked)

context code	n	DET	LAM
12	30	0.216	**0.083**
11	30	0.264	0.229
4	30	0.251	**0.265**
3	30	0.268	0.147
6	30	0.209	0.196
9	30	0.250	0.139

Kruskal-Wallis across contexts: DET H=1.74 **p=0.884** (NS) | LAM H=13.88 **p=0.016** (significant).

Verdict: graded bat bouts have **context-dependent LAMINARITY** — the degree to which the bat

holds a s 0.265

#2 Marmoset — token-free MP confirms degenerate inventory (tokenization-independent)

n=200 calls, raw 44-D fingerprint trajectory, **bypassing HECB + DUSTED entirely:**

recurren

Verdict: the degenerate-inventory claim (§11) is **tokenization-independent** — token-free MP on

raw signa an

#3 Bengalese — full transition-network topology (songbird-syntax lens)

n=100 bouts → **24 nodes**, **5 communities**, clustering **0.686**, density **0.475**, mean path

length **1

Verdict: **small-world + modular** — the topology songbird-syntax literature reports for Bengalese

finch son
without la

Structure discovery is now 100% complete. Three paradigms (alignment / graph / dynamics) agree on

the biolog

36. The Entanglement Ceiling — caller identity and call form are non-separably entangled (2026-06-23)

Question. §26 established that animal calls are *non-additively caller-bound* — the raw distance

ratio betw
meaning

Motivation. In human speech, CPC's contrastive objective partially disentangles speaker from word

content (
ethologica

Setup. Marmoset corpus (5 call types: Phee/Twitter/Trill/Tsik/Seep; 87 callers with ≥20 calls

each). Fo
caller labe

Module: `representation/disentangle.py` (pure numpy, firewall-clean, deterministic).

Method 1: Orthogonal Subspace Projection (LDA) — linear algebraic

Learns the caller-discriminating subspace via Linear Discriminant Analysis, then projects the

fingerprin
of

	caller V	form V	ratio	Δ caller	Δ form
Baseline (raw fp)	0.611	0.614	1.0	—	—
LDA projection	0.582	0.494	0.8	−0.029	**−0.120**

Verdict: FAILS. Form degrades 4× harder than caller (−0.120 vs −0.029). The caller and form

subspace

Method 2: Orthogonal Subspace Projection (LogReg) — linear algebraic

Same principle, but the caller-discriminating directions come from a multinomial Logistic Regression

weight ma

	caller V	form V	ratio	Δ caller	Δ form
Baseline (raw fp)	0.611	0.614	1.0	—	—
LogReg projection	0.606	0.601	1.0	−0.005	−0.013

Verdict: NO EFFECT. The LogReg weight matrix could not find a caller-discriminating subspace

separable

Method 3: Per-Speaker Z-Normalization — affine transform

For each caller, centers (subtracts mean) and scales (divides by std) every dimension across all

their calls than

	caller V	form V	ratio	Δ caller	Δ form
Baseline (raw fp)	0.611	0.614	1.0	—	—
Z-normalization	0.592	0.540	0.9	−0.019	**−0.074**

Verdict: FAILS. Form degrades 4× harder than caller (−0.074 vs −0.019). Per-speaker affine

normaliza entangler

Method 4: Delta-Trajectory — non-linear structural

Replaces absolute frame fingerprints with inter-frame differences: Δ_i = fp_{i+1} − fp_i. Theory:

static call

Tested on 200 marmoset calls → 4,623 frame windows (80 ms each, 44-D fp per frame) → 4,623 delta

vectors.

	caller V	form V	ratio	Δ caller	Δ form
Raw frame fp (absolute)	0.278	0.396	1.4	—	—
Delta trajectory (differenced)	0.173	0.233	1.4	−0.105	**−0.163**

Verdict: FAILS. The disentanglement ratio stays at 1.4 — zero improvement. Form degrades 1.5×

harder th spectral

Summary: the complete disentanglement proof

Method	Mathematics	Entanglement level tested	Δ caller	Δ form	Result
LDA projection	Linear subspace	Static spectral (linear)	−0.029	−0.120	**FAIL** (overlap)
LogReg projection	Linear weight	Static spectral (linear)	−0.005	−0.013	**FAIL** (inseparable)
Z-normalization	Affine (mean + scale)	Static spectral (affine)	−0.019	−0.074	**FAIL** (overlap)
Delta-trajectory	Non-linear differencing	Dynamic temporal (structural)	+0.105	−0.163	**FAIL** (temporal dynamics are

Conclusion. The 44-D acoustic fingerprint cannot be disentangled into pure caller identity and pure

call form
of

1. **Static spectral (linear):** the caller-discriminating subspace (LDA/LogReg) overlaps with the

call-type

The semantic ceiling is a mathematical fact, not a policy choice. In human speech, CPC's

contrastiv
from

Implications for the architecture:

The etho

AVES ablation — does a deep learned embedding break the ceiling? (reviewer-proofing)

A reviewer will argue the entanglement is an artifact of hand-crafted 44-D physics features, and that

a deep ne
directly:

Representation	Method	caller V	form V	ratio	Δ caller	Δ form	form drops ≥ caller?
fingerprint 44-D	baseline	0.276	0.552	2.00	—	—	—
fingerprint 44-D	LDA	0.251	0.459	1.83	−0.025	**−0.093**	YES (3.7×)
fingerprint 44-D	Z-norm	0.282	0.478	1.70	+0.006	**−0.074**	YES
AVES 768-D	baseline	0.226	0.628	2.78	—	—	—
AVES 768-D	LDA	0.227	0.631	2.77	+0.001	+0.003	no-op
AVES 768-D	Z-norm	0.212	0.587	2.77	−0.014	**−0.041**	YES (2.9×)

Honest verdict — deep learning narrows the ceiling but does not break it.

1. **AVES is a less-entangled representation than the fingerprint** (baseline ratio 2.78 vs 2.00): it

achieve
reviewer's

2. **But AVES is still not linearly disentangleable.** Z-normalization — the affine "scrub the caller"

transfor
form

3. **LDA is a no-op on AVES.** Projecting out the linear caller subspace from 768-D changes neither

caller V
(where

Conclusion. No firewall-compliant *available* representation — deterministic physics (44-D) or deep

SSL (768-

the gap

Reproducibility note. `entanglement_ceiling_aves.py` also reproduces §36's fingerprint {baseline, LDA,

Z-norm}
matches §

Named result: the Entanglement Ceiling

This is the **Entanglement Ceiling**: caller physical identity and call motor form are

non-linear

Biological interpretation — the medium is the message. The delta-trajectory failure is

the most

Human vs. animal — closing the loop. This sharpens the cross-species analogy and makes

the huma

Why this mandates Track C (the final pivot)

With this test, the computational representation space is officially exhausted. Because the computer

37. Compositional-structure mining — structural prerequisites only (Semantic-Ceiling-safe, 2026-06-24)

Question. Compositionality — a sequence AB carrying information/behaviour that A or B alone do n

Setup. Egyptian fruit bat, 2000 bouts. Sub-call segmentation = raw NBD (`bat_call_granular`, the §13/§3

T0 structural prerequisites detected:

Prerequisite	Evidence	Present?
Non-random transitions (PST)	1367 significant nodes, max-depth 4, ent	**YES** (rich syntax)
High-TP distinct bigrams	17→1 (ratio 18.7), 1↔11 reciprocal (coun	**YES**
Positional specialists	cluster 8 onset 0.68 (n=1222); clusters	**YES** (prefix/suffix signatures)
Transition-network topology	24 nodes, 5 communities, density 0.51	modular syntax

Cross-bout compositional-prerequisite test (honest negative). Does *bout-level occurrence* of the top alone?

Pattern	Cramér's V vs Context	n bouts
AB-present (17→1)	0.162 (p=0.005)	40
A-present (17)	0.195	84
B-present (1)	0.186	155

The AB↔Context association is **significant** (p=0.005) but does **NOT exceed** either part (0.162 <

T2 counterfactuals generated. For the top candidate bouts, the DSP Audio Splicer produced 25 compo

Positive control — LibriSpeech proves the pipeline *can* find compositional structure.

The bat §1

Verdict (ceiling-respecting). The bat repertoire exhibits strong *structural prerequisites*

for compo

Scope decision — no new corpus (v1.0 freeze respected). A within-bout animal "AB > parts"

test would

38. Compositional scan — zebra finch call sequences (Semantic-Ceiling-safe, 2026-06-25)

Question. Does the zebra finch **call** repertoire (not song) show compositional sequence

structure?

Setup. Adult corpus (2969 wavs, 221 sessions). Filename `<individual>_<date>-<VT>-<zz>.wav`;

canonical

Two flagged assumptions / fixes (honesty):

*Renditi

T0 structural prerequisites:

Prerequisite	Evidence	Present?
Non-random transitions (PST)	106 significant nodes, max-depth 4, entr	weak yes (predictable)
Transition-network topology	12 nodes, 75 edges, 9 communities, **den	**dense / near-uniform**
Sharp high-TP bigrams (common types)	none among Dc/Ne/Te/Th/Tu/Ag/Wh/So	**absent**
Positional specialists	only Di onset 0.53 (n=15, small)	**absent** (common types)
Individual-specificity (common bigram)	Dc→Te V=0.334, **p=0.428**	**absent**
Individual-specificity (rare bigram)	Be→Lt V=0.703, p=0.015	one bird (LblBlu) idiosyncrasy

The single high-TP distinct pair, **Be↔Lt (ratio 67, count 6), is one bird's (LblBlu)

idiosyncra

Verdict. Zebra finch calls are a **broad, freely-mixed repertoire**, not a compositional

syntax sy

39. Cross-corpus RQA — graded recurrent units / motif substrate (2026-06-25)

Question. Where do graded recurrent units / motifs live across repertoires? RQA is the

graded-dy

HONEST SCOPE. RQA characterises the graded recurrent **substrate** (recurrent units,

motif-like

Setup. 44-D frame trajectory per recording (WIN 80 ms / HOP 40 ms / ≤120 frames, matching

§34), `rqa

corpus	n	DET	LAM	motif_L	frames
Bengalese (song)	80	**0.364**	0.350	**2.42**	120

LibriSpeech (speech)	80	**0.352**	0.417	2.30	120
marmoset (calls)	80	0.330	0.325	2.00	27
zebra finch (calls)	27	0.227	0.100	2.00	14
bat (bouts)	80	0.067	0.000	1.00	14

Kruskal-Wallis across corpora: **DET p=1.2e-10**, **LAM p=2.5e-20** (graded-recurrent dynamics

differ stro

Verdict. **Bengalese song carries the strongest graded recurrent motifs** — DET 0.36,

motif_L 2.

Two flagged frame-count caveats (honesty). DET needs diagonal lines of length ≥2, which

requires e

40. Is "isolate individuals first" always necessary? A direct test (2026-06-25)

Question. A natural reading of §36 (Entanglement Ceiling) + §26 (rook cross-caller

retrieval b

Three discovery modes, measuring call-FORM recovery (Cramér's V of token ↔ call-type):

mode	V(token, call-type)	note
GROUP (pooled codebook, k=24)	**0.574** (p=0.005)	all callers pooled
PER-CALLER (per-bird k-means)	0.554	mean over 89 callers (k capped by n//8 —
CROSS-CALLER transfer	**0.552**	caller A's codebook → caller B's calls (

Decisive, unconfounded checks:

check	result	reading
P(nearest-neighbour same call-TYPE)	**0.626**	call-type dominates
P(nearest-neighbour same CALLER)	**0.029**	caller is weak at whole-call level
Group V(token, call-type) vs V(token, ca	0.574 vs **0.283**	form ≫ caller

Verdict — the recommendation is granularity-dependent, not universal (honest nuance). For

**coarse c

Where "isolate individuals first" DOES hold (so the recommendation survives, scoped):

fine-grain

Caveat (honesty): the per-caller V (0.554) is **underpowered**, not interpretable as

"per-calle

Synthesis. The Entanglement Ceiling (§36) and the "isolate individuals" recommendation are

both corre

41. Are the recurrent structures from individuals or across the group? (2026-06-25)

Question. When the pipeline finds recurrent structures (motifs / bigrams / units), are they

WITHIN-in

Method — individual-concentration vs a weighted-random null. For each structure (here,

call-type u

Result — recurrent BIGRAMS split ~50/50; recurrent UNITS are shared.

	recurrent bigrams (n=50)	
WITHIN-individual	**27 (54%)**	one bird's signature, beyond null
ACROSS-group	23 (46%)	shared species-level

Within-individual bigrams are strikingly concentrated — e.g. `Tu→So`/`So→Tu` (bird YelOra,

H=1.000)
the

Unigram (call-type) attribution is the inverse: the **common** call types (Dc/Te/Ne/Th/Tu/Ag/

Wh/So) ar

Verdict — the resolution of the §36/§40 tension. The **underlying UNITS** (call types) are

shared a
(bigrams

42. Recurrent-structure attribution across corpora (2026-06-25)

Question. Does the §41 within-vs-across split generalise beyond zebra finch? The

attributio

corpus	n_rec	n_ind	#bigrams	%WITHIN-individual	top within
Bengalese (song, 1,379 syllables)	1,379	11	92	**84%**	`7->7` (Bird0, top1 1.00)
zebra finch (calls)	173	24	50	**54%**	`Tu->Te` (GraGra, top1 1.00)
marmoset (calls)	74	74	25	**0%**	— (see caveat)
LibriSpeech (human words)	603	13	329	**26%** (74% across)	share speaker-specific phrases

Verdict — the §36/§41 prediction is confirmed across repertoires. The % of recurrent

structure

Bengalese song = 84% within-individual — the strongest. Each bird has its own syllable

motif gra

Synthesis. Animal recurrent structure (song most, calls next) is substantially

**individu
whereas

43. RQA on RookID — corvid field calls (2026-06-26)

Question. Rook was deferred from the §39 cross-corpus RQA table (its ~2 h 4-channel field wavs can'

Result (honest — dominated by a frame-count artifact).

source	n	DET	LAM	dur_s
Balbo	71	0.000	0.000	0.47
Feisty	71	0.000	0.000	0.52
Kafka	53	0.000	0.000	0.44
Brain	47	0.000	0.000	0.46
Jolene	44	0.000	0.000	0.51
Gigi	26	0.333	0.250	0.56

Corpus-level DET=0.000, LAM=0.000. Kruskal-Wallis across sources: DET p=0.024, LAM p=0.21.

Verdict (read carefully, do not overclaim).

1. **Rook

Completion of the §39 table. Rook confirms the pattern: short isolated calls (rook, bat, zebra fin speech,